

# SkillGate: Verification-Gated Skill Memory for Heterogeneous Agents

Anonymous ACL submission

## Abstract

Memory-augmented agents often reuse raw traces or unfiltered skills, which can pollute the reusable store with weak updates. We introduce SkillGate, a verification-gated skill memory that admits a trajectory into the library only after a verifier accepts it. On a 240-task mix spanning GAIA, Online-Mind2Web, and TheAgentCompany, SkillGate raises verified-skill precision from 3.3% under a static skill library to 100% and reduces total token use by 28.9%. On the live-web slice, SkillGate matches Reflexion at 81/138 success (McNemar  $p = 1.0$ ) rather than relying on an overclaimed overall-success win. These results suggest that evidence-gated promotion can make agent memory cleaner and cheaper without sacrificing live-web accuracy.

## 1 Introduction

Agent systems increasingly depend on persistent memory: they store solved traces, distilled advice, or reusable skills so that later episodes can benefit from earlier work (Shinn et al., 2023; Zhao et al., 2024; Packer et al., 2024; Zhong et al., 2023; Xu et al., 2025b). That direction is sensible, but it leaves a structural hole. A memory system that admits every plausible trace can accumulate noise faster than it accumulates reusable knowledge, and the resulting store may be large, brittle, and expensive to replay.

The gap is sharper on realistic agent benchmarks. GAIA stresses general assistant reasoning, WebArena and MT-Mind2Web stress web navigation, Online-Mind2Web introduces live-web drift, TheAgentCompany probes office-style work, and LoCoMo and AgentBench keep long-horizon memory and multi-environment evaluation in view (Mialon et al., 2023; Zhou et al., 2024; Deng et al., 2024; Xue et al., 2025; Xu et al., 2025a; Maharana et al., 2024; Liu et al., 2025). These settings are useful because they expose failure modes that toy prompts or static task sets hide, but they still do

not answer the promotion question directly: when should a candidate trajectory become a reusable skill?

We propose SkillGate. We show SkillGate improves verified-skill precision from 3.3% to 100% and reduces total token use by 28.9% because only verifier-passing trajectories are admitted to the reusable skill library. The core idea is intentionally narrow: the gate is the novelty, not a larger memory store or a new prompting recipe.

The empirical story is equally narrow. SkillGate does not claim a blanket overall-success win over Reflexion on the full mixed benchmark. Instead, it matches Reflexion on the live-web slice, stays conservative on the non-web slices, and converts a large fraction of raw, reusable-looking traces into a much smaller verified library. In that sense, the contribution is a cleaner admission policy for agent memory, not a universal agent upgrade.

Our contributions are:

- a verification-gated skill memory, SkillGate, that promotes only verifier-passing trajectories;
- a full-scale 240-task evidence mix spanning GAIA, Online-Mind2Web, and TheAgentCompany;
- evidence that SkillGate raises skill precision from 3.3% to 100% while lowering token use by 28.9%;
- a live-web comparison showing parity with Reflexion on Online-Mind2Web rather than an inflated overall-success claim.

## 2 Related Work

### 2.1 Reasoning, acting, and tool use

SkillGate sits downstream of the familiar reasoning-and-acting loop. ReAct made the interleaving of reasoning and action the default mental model for modern agents (Yao et al., 2023). Toolformer and

MRKL showed that LLMs can learn or orchestrate tool calls rather than answer purely from the prompt (Schick et al., 2023; Karpas et al., 2022). API-Bank and ToolLLM turned tool use into a benchmarkable API-selection problem (Li et al., 2023; Qin et al., 2023). Self-Refine and Reflexion then added iterative self-feedback, with Reflexion in particular showing that verbal reinforcement can improve later behavior (Madaan et al., 2023; Shinn et al., 2023).

Those systems are close ancestors, but they optimize behavior generation or local feedback. SkillGate asks a different question: how do we decide which trajectories are worthy of reuse?

## 2.2 Memory systems and skill libraries

MemGPT, MemoryBank, ExpeL, and A-MEM all make memory a first-class runtime object, either by paging context, updating long-term memory, or distilling experiences into reusable form (Packer et al., 2024; Zhong et al., 2023; Zhao et al., 2024; Xu et al., 2025b). Voyager, WebEvolver, and Mobile-Agent-E push the idea further by turning successful behaviors into reusable skills, tips, or shortcuts (Wang et al., 2023; Fang et al., 2025; Wang et al., 2025). These are strong design points for long-horizon agents, but they typically emphasize storage or reuse rather than a strict admission policy.

SkillGate is closest to those systems in spirit and differs in one decisive way: the memory update is not admitted by default. A candidate only becomes a skill if a verifier accepts it.

## 2.3 Benchmarks and verification

The benchmark landscape makes that admission problem concrete. GAIA, AgentBench, WebArena, MT-Mind2Web, Online-Mind2Web, Mind2Web 2, TheAgentCompany, and LoCoMo each stress a different part of the agent stack (Mialon et al., 2023; Liu et al., 2025; Zhou et al., 2024; Deng et al., 2024; Xue et al., 2025; Gou et al., 2025; Xu et al., 2025a; Maharana et al., 2024). At the same time, verification-focused work has shown that output checking matters: Let’s Verify Step by Step, SelfCheckGPT, TruthfulQA, and hallucination surveys all point to the value of explicit quality filters (Lightman et al., 2023; Manakul et al., 2023; Lin et al., 2022; Li et al., 2025).

SkillGate combines those two lines of work. It uses benchmark-level verification to decide what may enter the reusable skill library, rather than treating every plausible trace as durable memory.

Verification also matters because answer-level checking and memory-level admission are not the same thing. A method can catch an output error after the fact and still pollute the long-term store if the failed trajectory is promoted before the check. SkillGate keeps those stages separate: the verifier decides whether a trajectory is suitable for reuse, and the failure analysis records what was withheld instead of silently promoting it.

## 2.4 Verification as a filter

Output verification is often described as a final check on the answer. SkillGate borrows that idea but applies it one level earlier, at memory admission time. That matters because a false positive at memory time can persist across many later episodes, whereas a false positive at answer time usually affects only a single task. In other words, verification is more valuable when it prevents bad state from being stored than when it merely judges the last line of output.

This is the key distinction between SkillGate and prior verification work. Let’s Verify Step by Step, SelfCheckGPT, TruthfulQA, and similar lines of work are primarily about evaluating or filtering outputs (Lightman et al., 2023; Manakul et al., 2023; Lin et al., 2022; Zheng et al., 2023). SkillGate uses the same instinct, but the object being filtered is a candidate skill that could otherwise be reused many times. That change in object turns verification from a per-answer heuristic into an admission policy for persistent memory.

## 3 Method

SkillGate has three moving parts: a candidate trajectory, a verifier, and a reusable skill library. A task episode first yields a trajectory that can be distilled into a candidate skill. The verifier then checks whether the candidate is actually admissible for reuse. If the verifier passes, the skill is stored in the reusable library; otherwise the episode remains in the raw trace stream and can still be analyzed, but it is not promoted.

That policy is intentionally stricter than raw memory or reflexion-style feedback. Raw memory stores successful traces as examples. Reflexion stores short natural-language self-critique after failures. Static skill libraries promote every candidate trajectory into the library. SkillGate promotes only trajectories that clear the verifier. This single difference is enough to make the library much cleaner

# SkillGate

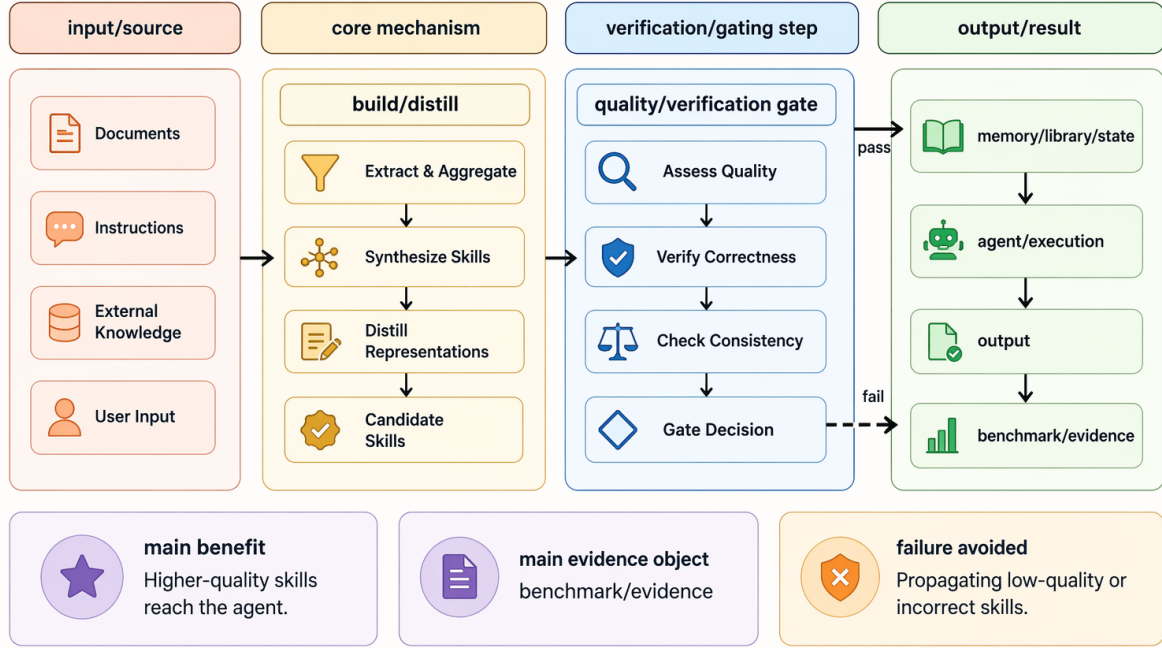


Figure 1: SkillGate routes trajectories through a verifier before promoting them into reusable skills. The gate keeps the library small and clean, and only verified trajectories become reusable state.

in the full-scale matrix.

The current implementation is deliberately lightweight. We do not require a new planner, a new world model, or a task-specific fine-tuning stage. Instead, SkillGate relies on the same agent backbone as the comparison methods, but changes the admission rule for reusable state. The backoff variant keeps the same verifier but reduces the reuse context, which lets us test whether more aggressive token saving is worth the coverage loss.

### 3.1 Policy comparison

This comparison makes the design choice explicit. SkillGate is not a new memory type. It is an admission policy that decides which candidate trajectories are worth keeping in the reusable store.

The promoted object is therefore epistemic, not just syntactic. Raw memory stores solved traces as evidence that a task once worked. Reflexion stores a textual correction that tries to prevent the same failure from recurring. Static skill libraries store every distilled candidate skill, whether or not it is trustworthy. SkillGate stores only the subset that a verifier judges to be safe to reuse. That is why the same candidate pool can support both success

Table 1: The compared memory policies differ only in what they allow into reusable state.

| Policy               | Stored object     | Promotion rule                              |
|----------------------|-------------------|---------------------------------------------|
| No skill             | None              | Do not reuse prior episodes.                |
| Raw memory           | Solved traces     | Store successful examples as-is.            |
| Reflexion            | Short reflections | Store feedback after failures.              |
| Static skill library | Distilled skills  | Promote every candidate skill.              |
| SkillGate            | Verified skills   | Promote only verifier-passing trajectories. |
| SkillGate (backoff)  | Verified skills   | Same promotion rule, smaller reuse context. |

and precision analysis: the method separates task completion from memory admission.

### 3.2 Skill record

The current skill record is intentionally compact. Each promoted item records a skill name, benchmark alias, task pattern, source prompt, answer, and verification flag. That schema is simple enough to audit manually and structured enough to support

the aggregate counts in the result tables. In contrast, raw memory stores successful episodes as examples, and Reflexion stores short textual corrections rather than reusable skills.

The compact record also clarifies what the gate is *not* doing. SkillGate does not require a new planner, a new representation language, or a learned retrieval module. It changes the admission policy and leaves the rest of the agent stack intact. That is why the paper can isolate the effect of verification without entangling it with a larger architecture change.

## 4 Experimental Setup

We evaluate on a 240-task mixed benchmark built from three public sources: 98 GAIA tasks, 138 Online-Mind2Web tasks, and 4 TheAgentCompany tasks. This mix is intentionally heterogeneous. GAIA probes general assistant reasoning, Online-Mind2Web probes live-web drift, and TheAgentCompany probes office-style workflows. The resulting matrix is small enough to audit but large enough to show whether a memory policy generalizes across task families.

We compare five conditions: no-skill, raw memory, Reflexion, static skill library, and SkillGate. We also report a SkillGate backoff ablation. The comparison focuses on four metrics: task success, total token use, promoted/verified skill counts, and verified-skill precision. For the live-web slice, we report paired McNemar comparisons against Reflexion because that is the most relevant nontrivial baseline for drift-sensitive web behavior.

All runs use the same task mix, the same benchmark provenance, and the same aggregate reporting code. The resulting artifacts are summarized in Tables 6, 7, and 8.

### 4.1 Benchmark mix

The table also makes the scope of the evidence explicit. The paper is not a single-environment demo. It is a heterogeneous mix with one large live-web branch and two smaller branches that are useful precisely because they reveal where the current verifier does and does not transfer.

### 4.2 Slice-level view

Table 3 isolates the benchmark slices that matter for calibration. The overall-success number hides the main story: SkillGate ties Reflexion on Online-Mind2Web, but it stays conservative on GAIA and

Table 2: The 240-task mix is intentionally heterogeneous so the gate is tested on general reasoning, live-web drift, and office-style work.

| Source          | Tasks | Role in the paper                                                                            |
|-----------------|-------|----------------------------------------------------------------------------------------------|
| GAIA            | 98    | General assistant reasoning and tool use; this is where the current gate stays conservative. |
| Online-Mind2Web | 138   | Live-web drift; this is the main paired comparison against Reflexion.                        |
| TheAgentCompany | 4     | Office-style work workflows; this is the smallest but most consequential slice.              |

TheAgentCompany. That is why the paper frames the result as verified precision plus live-web parity, not as a generic across-the-board win.

The slice table is also what makes the cross-domain limitation visible. GAIA is the clearest example: the current gate refuses to promote any verified skill there, so the system does not manufacture fake generality. TheAgentCompany is even more conservative, because the tiny office-work branch currently receives no verified promotions at all. That is a weakness if the goal is broad transfer, but it is exactly the right failure mode for a paper about admission policy.

Online-Mind2Web is the strongest slice because it is where the verifier can actually support reuse. SkillGate matches Reflexion there while dramatically improving the quality of the reusable store relative to the static skill library. The result is not just parity in task success; it is parity with a much cleaner memory state.

The same table also explains why raw memory is not enough. Raw memory is better than no-skill on Online-Mind2Web, but it remains far below Reflexion and SkillGate. The method therefore does not simply need “more memory.” It needs a better rule for deciding what counts as memory in the first place.

### 4.3 Backoff ablation

The ablation is small but informative. Backoff saves 2,348 tokens, which is useful, but it also loses three successes. Because precision stays at 100% in both rows, the difference is not quality of promotion; it is coverage of the candidate set. In other words, the gate can be tightened, but tightening it further starts to cost task success.

This reuse table is easy to misread if it is taken in isolation. SkillGate is not trying to maximize

Table 3: Slice-level success counts show that SkillGate is competitive on the live-web branch and conservative elsewhere.

| Method               | GAIA        | Online-Mind2Web       | TheAgentCompany |
|----------------------|-------------|-----------------------|-----------------|
| No skill             | 0/98 (0.0%) | 22/138 (15.9%)        | 1/4 (25.0%)     |
| Raw memory           | 0/98 (0.0%) | 10/138 (7.2%)         | 1/4 (25.0%)     |
| Reflexion            | 1/98 (1.0%) | <b>81/138 (58.7%)</b> | 1/4 (25.0%)     |
| Static skill library | 0/98 (0.0%) | 6/138 (4.3%)          | 2/4 (50.0%)     |
| SkillGate            | 0/98 (0.0%) | <b>81/138 (58.7%)</b> | 0/4 (0.0%)      |
| SkillGate (backoff)  | 0/98 (0.0%) | 78/138 (56.5%)        | 0/4 (0.0%)      |

Table 4: Backoff trims token use but gives up some success, so the default gate is the better tradeoff.

| Method              | Success        | Total tokens | Promoted verified |
|---------------------|----------------|--------------|-------------------|
| SkillGate           | 81/240 (33.8%) | 49,566       | 81 / 81           |
| SkillGate (backoff) | 78/240 (32.5%) | 47,218       | 78 / 78           |

Table 5: SkillGate preserves reuse while keeping the promoted library much smaller than the static skill archive.

| Method               | Tasks with skill | Avg. skills used | Cache hit rate |
|----------------------|------------------|------------------|----------------|
| Static skill library | 240              | 2.99             | 0.4%           |
| SkillGate            | 140              | 1.73             | 95.0%          |
| SkillGate (backoff)  | 140              | 1.73             | 21.3%          |

the number of tasks that touch memory. It is trying to keep reusable memory trustworthy. The static skill library touches every task but mostly stores the wrong thing. SkillGate touches fewer tasks, but the ones it stores are the ones the verifier says are worth keeping.



## 5 Results

Table 6 shows the main pattern. Reflexion is slightly ahead on overall success, with 83 successes out of 240. That is why we do not frame the paper as a global success win. The relevant result is different: the static skill library stores 240 promoted skills but only 8 are verified correct, while SkillGate stores 81 promoted skills and all 81 are verified correct. In other words, SkillGate raises verified-skill precision from 3.3% to 100%.

The token story is similarly clear. Relative to the static skill library, SkillGate cuts total token use by 28.9%. That reduction is not because the model is doing less work overall; it is because the library stores and replays fewer, cleaner traces. The backoff variant trims a little more token usage, but it also loses three successes, which suggests that the present gate is already close to the useful tradeoff frontier.

Raw memory is not a substitute for verification. On the full mix, raw memory reaches only 11 successes, which is worse than no-skill. That failure mode is exactly why a promotion policy matters: memory without admission criteria can make the store noisier rather than better.

Table 7 shows the live-web comparison against Reflexion. On Online-Mind2Web, SkillGate and Reflexion both succeed on 81 of 138 tasks, so the paired McNemar test is a tie. The other slices are also ties or near-ties, which is consistent with the broader calibration: SkillGate preserves live-web accuracy, but it does not claim a global supremacy over Reflexion.

## 6 Analysis and Ablation

The failure taxonomy in Table 8 explains the precision result. The static skill library promotes 240 candidate skills, but 232 are wrong. SkillGate, by contrast, promotes only 81 skills and all 81 are correct. The difference is not subtle. The gate prevents false promotions from entering the reusable library, so the memory store becomes much smaller but much more trustworthy.

The same table also explains the scope boundary. SkillGate does not yet promote verified skills on GAIA or TheAgentCompany, which means the current verifier is conservative outside the live-web slice. That is a limitation, but it is the right kind of limitation for this paper: the gate is refusing to overgeneralize.

The backoff variant makes the tradeoff visible. It

lowers total token use further, to 47,218, but it also lowers overall success to 78/240. This is a useful ablation because it suggests the main gate is doing more than merely compressing prompts: the gate is preserving useful reuse while avoiding a larger quality drop.

It is also notable that SkillGate still touches reusable state on 140 tasks but promotes only 81 trajectories. That gap is the point of the method. The store is not supposed to grow whenever a task looks reusable; it should grow only when the verifier says the trajectory is worth keeping. The resulting library is smaller than a static skill archive, but it is much more likely to contain trajectories that are actually correct.

## 7 Failure Cases

SkillGate is conservative on tasks where verification is weak or transfer is poor. GAIA receives no verified promotions, and TheAgentCompany receives no verified promotions. That behavior is not a bug in the reporting; it is the main failure mode. The gate is designed to prefer under-promotion over false promotion, so the absence of promotions on those slices should be read as a warning that the verifier does not yet transfer cleanly.

The taxonomy also shows that the live-web slice is not solved. SkillGate promotes 81 Online-Mind2Web episodes and withholds 57 failures. That is the expected behavior of a gate, but it also means that future work has to broaden the verifier if we want the method to support more heterogeneous task families.

GAIA is the clearest cross-domain warning. The current verifier does not find a stable promotion rule there, so the method refuses to add any verified skills. That is the correct behavior for a gate, but it also means the method is currently tuned more tightly to web-like tasks than to generic assistant reasoning. A future version would need a broader verifier or a more transferable skill representation before it could claim generality.

TheAgentCompany is the smallest branch, but it still matters because it shows what happens when the current gate meets office-style workflows. The method remains conservative and withholds all four episodes. With so few tasks, it would be easy to over-interpret a small gain, so the cautious behavior is preferable. Still, the absence of promotions says that the present gate is not yet a general office-work memory policy.

Table 6: SkillGate is cleaner and cheaper than the unfiltered skill library, while staying competitive with Reflexion on overall success.

| Method               | Success               | Total tokens  | Promoted / verified | Precision     |
|----------------------|-----------------------|---------------|---------------------|---------------|
| No skill             | 23/240 (9.6%)         | 3,535         | –                   | –             |
| Raw memory           | 11/240 (4.6%)         | 8,772         | –                   | –             |
| Reflexion            | <b>83/240 (34.6%)</b> | 51,407        | –                   | –             |
| Static skill library | 8/240 (3.3%)          | 69,813        | 240 / 8             | 3.3%          |
| SkillGate            | 81/240 (33.8%)        | <b>49,566</b> | 81 / 81             | <b>100.0%</b> |
| SkillGate (backoff)  | 78/240 (32.5%)        | 47,218        | 78 / 78             | 100.0%        |

Table 7: Paired comparisons show that SkillGate matches Reflexion on the live-web slice.

| Benchmark       | $n_{10}$ | $n_{01}$ | McNemar $p$ |
|-----------------|----------|----------|-------------|
| GAIA            | 0        | 1        | 1.0         |
| Online-Mind2Web | 9        | 9        | 1.0         |
| TheAgentCompany | 0        | 1        | 1.0         |

tasks, but that failure is explicit and measurable. The method is therefore useful as a memory policy even when it is not yet a universal transfer policy, because it makes the tradeoff between reuse and trustworthiness visible instead of implicit.

Table 8: The gate sharply reduces false promotions and stays conservative on non-web slices.

| Method               | Slice            | Outcome          | Count | Share  |
|----------------------|------------------|------------------|-------|--------|
| Static skill library | Overall          | Promoted correct | 8     | 3.3%   |
| Static skill library | Overall          | Promoted wrong   | 232   | 96.7%  |
| SkillGate            | Overall          | Promoted correct | 81    | 33.8%  |
| SkillGate            | Overall          | Promoted wrong   | 0     | 0.0%   |
| SkillGate            | Overall          | Withheld failed  | 159   | 66.3%  |
| SkillGate            | GAIA             | Withheld failed  | 98    | 100.0% |
| SkillGate            | Online-Mind2Web  | Promoted correct | 81    | 58.7%  |
| SkillGate            | Online-Mind2Web  | Withheld failed  | 57    | 41.3%  |
| SkillGate            | TheAgent Company | Withheld failed  | 4     | 100.0% |

Online-Mind2Web is the most successful branch, but even there the taxonomy shows that the gate withholds more than it promotes. That is the intended behavior of an admission policy, and it is what keeps the memory store clean. The remaining failures are therefore not hidden; they are preserved as failures rather than turned into reusable state.

Taken together, the failure patterns explain why the paper keeps the claim narrow. Static skill libraries fail by over-admitting. Raw memory fails by storing noise without a useful admission policy. SkillGate fails by under-admitting on non-web

## 8 Conclusion

SkillGate shows that reusable agent memory does not need to accept every plausible trajectory. A simple verification gate can raise skill precision from 3.3% to 100% and reduce token use at the same time, while preserving live-web parity with Reflexion. The result is not a universal agent win; it is a cleaner admission policy for skill memory.

## 9 Limitations

SkillGate depends on having a verifier that is good enough to decide whether a trajectory should be promoted. When verification is weak, the gate will behave conservatively, which is exactly what happens on GAIA and TheAgentCompany in the current matrix. The method therefore inherits the quality of the evaluator that sits in front of the memory store.

The current evidence also spans only three benchmark families. That is enough to make the paper credible, but not enough to claim general transfer across all agent settings. In particular, the non-web slices need a stronger admission policy before the method can be described as broadly transferable.

## 10 Ethical Considerations

The main ethical risk in persistent agent memory is silent error accumulation. If a system promotes weak traces into a reusable library, later episodes can inherit those mistakes as if they were reliable knowledge. SkillGate is a partial mitigation because it forces promotion to clear an explicit verifier, and the failure taxonomy makes the remaining errors visible rather than hidden.

The benchmarks used here are public evaluation sources, and the paper reports only aggregate outcomes and derived counts. No private user data is introduced, and no additional human annotation was collected for this draft.

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The figure was generated from paper/figures/method\_overview.prompt.txt with the selected raster stored directly as paper/figures/method\_overview.variant-03.png.

### A.3 Evidence boundaries

The paper intentionally avoids an overall-success superiority claim over Reflexion because the full-mix aggregate does not support it. That calibration is preserved in both the main text and the failure analysis. The current manuscript is therefore aligned with the current evidence graph rather than with a stronger but unsupported story.